

Bell Nozzle Rocket Engine Calculations

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Nomenclature

C_f	Vacuum thrust coefficient
$\dot{m}$	Mass flow rate
M	Molar mass
Ma	Mach number
$mol\%$	Mole percentage
P	Pressure
R	Gas constant
T	Temperature
v	Velocity
ϵ	Nozzle area ratio
γ	Ratio of specific heats
ρ	Density

Subscripts

a	Ambient
c	Chamber
e	Exit
s	Stagnation
t	Throat

Introduction

Documentation of calculations for a bell nozzle rocket engine

1 Propellant

1.1 Propellant Stoichiometry

2 Exhaust Gasses

2.1 Ratio of Specific Heats of a Multi-Compound Gas Mixtures [1]

$$\frac{1}{\gamma-1} = \sum \frac{mol\%}{\gamma_i-1}$$

2.2 Stagnation Pressure of Exhaust Gasses

$$P_s = P_1 \left(1 + \left(\frac{\gamma-1}{2} \right) Ma_1^2 \right)^{\frac{\gamma}{\gamma-1}}$$

2.3 Stagnation Temperature

3 Combustion Chamber Geometry

4 Throat Geometry

5 Nozzle Geometry

[1]

Bibliography

References

- [1] O. Lanzi, "How do you calculate the heat capacity ratio for a multi-compound gas?." <https://chemistry.stackexchange.com/a/166796>.